Fiscal Year:	FY 2012	**Task Last Updated:**	FY 02/08/2013
PI Name:	Kim, Jungho Ph.D.		
Project Title:	Development and Validation of Design Tools for Advanced, Two Phase, Space Heat Exchangers		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	20742-0001	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	05/19/2009	**End Date:**	08/18/2012
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 8/18/2012 per NSSC info (Ed., 6/02/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Dessiatoun, Serguei (University of Maryland)		
Grant/Contract No.:	NNX09AK39A		
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Task Description:	The proposed research will develop the data and models needed to predict the behavior of two-phase flows in geometries relevant to advanced heat exchangers in variable gravity environments. Unlike previous work where only time- or space-averaged heat transfer data were measured, we will obtain local measurements of the wall heat transfer coefficient with high temporal and spatial resolution. In the proposed effort, an infrared video (IR) camera will be used to obtain temperature distributions along the inside and outside walls of a heated silicon tube so local heat transfer coefficients can be determined. Inverse heat conduction techniques can be used to obtain transient heat transfer coefficients as the bubbles pass. Visualization of the flow will allow the heat transfer to be correlated with the flow regime. A test rig will be constructed to enable measurements of two-phase flow boiling heat transfer in variable gravity environments. Results of the proposed work can serve as benchmarks to validate analytical and/or numerical codes, and will impact the ability to reliably engineer the energy rich systems for future space missions. Although this proposal is primarily aimed at		

	thermal management subsystems and development of lightweight evaporators, it also impacts other systems for novel power and transmission, advanced chemical propulsion, and energy storage.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed research will clarify the role of gravity in flow boiling heat transfer so better predictions can be made of boiling in earth gravity environments as well as in the microgravity of space. The novel techniques developed as part of this research are also applicable to the study of multiphase phenomena in other complex geometries as well.
Task Progress:	The work over the past three years focused on the following areas, test apparatus construction and qualification, development of an IR thermometry technique, and testing of heat transfer in variable gravity environments. Progress in each area is summarized below. Construction of flight test apparatus Between March–September, 2010, effort was focused on the design and construction of a flow loop that could be flown on low-g aircraft. The silicon tube was 6 mm ID with a 1 mm thick wall, and 10 cm long. The voltage required to produce a heat flux of 4 W/cm2 in this tube was 300 V. Visualization of upward flow in the silicon tube was obtained using hexane as a working fluid for preliminary experiments in the lab. The test apparatus was documented in a Test Equipment Data Package and submitted to NASA. Upon acceptance, the test apparatus was shipped to Ellington Field and flown on the Zero-G Corporation aircraft. It was observed that in Earth gravity or high gravity, the liquid experienced a body force that overcame the vapor pressure within the film, enabling the liquid to wet the wall for a longer distance from the inlet. In low-g, this body force does not exist so the liquid moved off the surface more easily and inverted annular flow occurred at lower qualities and heat fluxes than in high-g. In the lower end of the tube, bubbly flow was visible, but the flow was observed to detach about 25% up from the inlet of the tube. In addition to inverted annular flow, bubbly flow and dryout were observed. Since the maximum flow rate that was performed at both days (0.35m/s) was relatively low, changes in bubble size and bubble behavior were clearly noticeable between the transitions from high to zero-g. Some problems were identified with the test apparatus during the initial flights. For example, the flow within the test section was observed to be quite unstable, and the size of the ice bath tank was unnecessarily large. Modifications to the test apparatus was performed to alleviate these problems for the follow-on flights in May-June, 2011. A mirror array in the test section was installed so that the full temperature distribution on one half of the silicon tube could be obtained simultaneously with unobstructed visualization of the flow. IR Thermometry Although the IR camera has been used until now to perform qualitative visualization of the fluid flow within a silicon tube, it is ultimately desired to obtain quantitative measurements of the wall temperature along the tube so the local heat flux can be measured. In order to perform these measurements, the wetted surface must be coated with a black coating so its temperature can be measured, and the temperature variations produced by passing bubbles must be large enough to be measurable. Because silicon is highly conductive (k=120 W/m-K), the temperature variations produced by passing bubbles are very small. In order produce a measurable temperature variation, it was decided to coat the inside of the tube with a thin, low thermal conductivity film onto which a black coating could be applied. The film needed to be thin enough so the temperature across it was not excessive, yet thick enough to produce the temperature variations needed for the IR measurements. Consider a polyester (Mylar) film 0.004 mil (100 microns) thick. Since the thermal conductivity of polyester is 0.15 W/m-K, a heat flux of just 5 W/cm2 would produce a large, 33°C temperature drop across the film. This low conductivity, however, does allow very sharp spatial changes in heat transfer to be resolved. A problem that arises when applying a non-transparent film to the inside of the tube is that the film itself will emit and absorb radiation. The total radiation detected by the IR camera is the sum of the energy emitted by the black coating that has not been absorbed by the film and the energy emitted by the film itself. Since passing bubbles will result in a time varying temperature profile within the film, the radiant energy emitted by film will also change. To address these problems and identify an appropriate material, the absorption coefficient of various materials were measured. Polyester and polyimide were found to be partially transparent to IR with absorption coefficients of 7500 m-1 and 8800 m-1, respectively (in contrast, the coefficient for silicon was about 52 m-1), so a 4 mil (100 micron) thick layer of polyester would transmit only about 48% of the incident energy through it. Based on these measurements, and based on numerical simulations of the temperature variations that would be produced by expected changes in the heat transfer coefficient produced by passing bubbles, it was decided to use a 0.5 mil or 1.0 mil thick polyimide layer with 1.0 mil silicone adhesive as the film on the inside of the silicon tube. To address the problem of radiation absorption within the film and the non-uniform temperature distribution, an algorithm was developed whereby the temperature of the black coating along with the instantaneous temperature profile within the polyimide film could be determined. In this algorithm, the radiant energy recorded by the IR camera was used along with a guessed initial temperature to extract a corrected black coating temperature. This temperature was then set as a boundary condition for a 1-D transient simulation of the temperature within the film. Although the temperature distribution within the film was incorrect at small times due to the guessed initial temperature profile, the initial transient died out quickly and the true black coating temperature could be obtained. The algorithm was tested using a sample set of data. Details of the IR thermometry technique were provided in a journal paper: Kim, T.H., Kommer, E., Dessiatoun, S., and Kim, J., “Measurement of Two-Phase Flow and Heat Transfer Parameters Using Infrared Thermometry”, International Journal of Multiphase Flow, 2012, Vol. 40, pp. 56-67, doi: 10.1016/j.ijmultiphaseflow.2011.11.012. Experimental Results Over two flight campaigns, over 80 zero gravity parabolas were used to gather data in both low-g and 1.8 g environments. Before measurements were obtained, the fluid volume of PF-5060 was degassed and filtered. A vacuum pump drawing on the vapor side of the liquid-vapor membrane filter was run for a period of 6 hours while the fluid loop was circulated at a moderate flow rate (500 ml/min). At hour intervals, the degassing was stopped, and total gas measurements were performed. The total gas within the PF-5060 could be measured by pulling a vacuum on the air side of the accumulator, thus lowering the pressure of the fluid. The absolute pressure and fluid temperature were recorded

	and the total gas was calculated using Henry's Law. Six hours of continuous degassing reduced air saturated PF-5060 from a total gas reading of 1500 ppm to less than 5 ppm. IR video was obtained at a frame rate of either 50 or 200 Hz. After the data analysis algorithm was used to determine the true inside and outside wall temperatures, as well as local heat flux and convection coefficient. At flow rates above 50 kg/m2-s there is no appreciable difference between the zero and high gravity levels. It appears the effects of gravity on heat transfer is only significant at lower mass flux levels even though the flow regime changes. The reasons for this are currently being investigated in follow-on work.
Bibliography Type:	Description: (Last Updated: 03/13/2026)
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